

Ching Him Leung

☎ (217)693-1360 • ✉ himhugo@gmail.com • 🌐 hugo-leung.github.io
🆔 0000-0001-7907-3728 • in Ching Him Leung

Summary

Data-focused physicist with experience analyzing large-scale experimental datasets, building Monte Carlo models, and designing statistical inference pipelines. Skilled in Python, C++, SQL, Linux, HPC workflows, and containerized computing. Experienced in converting noisy real-world measurements into validated quantitative insights.

Technical Skills

Programming: Python, C/C++, SQL

Data Analysis: NumPy, pandas, SciPy, ROOT

Modeling: Monte Carlo methods, statistical analysis, **Systems:** Git, Apptainer, Linux, HPC regression, uncertainty quantification

Experience

Thomas Jefferson National Accelerator Facility

Hall C Postdoctoral Fellow

2024 – Present

- Performed data-quality checks and preliminary analysis for GEM-based tracking detectors used in high-rate experiments.
- Conducted detector setup, calibration, and data-quality checks for the LAD experiment in Hall C.
- Conducted simulation and feasibility studies to evaluate expected detector performance for planned measurements.
- Led testing of micropattern gaseous detectors to assess behavior under high-rate operating conditions.

University of Illinois Urbana-Champaign

Research Assistant

2019 – 2024

- Analyzed large datasets for Drell–Yan and charmonium studies, extracting ratios and distributions using optimized statistical workflows.
- Implemented uncertainty-aware inference pipelines integrating Monte Carlo variance and simulation corrections.
- Modified and validated Monte Carlo generators by enforcing physical kinematic constraints.
- Engineered reproducible analysis environments using Linux, Git, and Apptainer.
- Co-authored multiple peer-reviewed publications, including work published in *Phys. Rev. Lett.*, *Phys. Lett. B*, and *Nature*.

Education

University of Illinois Urbana-Champaign

Ph.D. in Physics

2018 – 2024

Thesis: Probing Parton Distributions in Proton using Drell-Yan and Charmonium Production with 120 GeV Proton Beam at Fermilab.

Teaching: TA for Modern Experimental Physics, Classical Mechanics II, and Atomic Physics & Quantum Theory.

The Chinese University of Hong Kong

B.S. in Physics

2014 – 2018

Awards

University of Illinois Urbana-Champaign

Felix T. Adler Award

2022

Recognized for excellence in graduate nuclear physics research.

University of Illinois Urbana-Champaign

Maurice Goldhaber Scholar Award

2020

Awarded for outstanding performance in an intensive graduate physics research program.